Place, pathways and regional cost context

Public climate source areas · two pathways · explicit geography and model coverage · 22 October 2026

Read the source area before the number

The explorer accounts for all 575 Nations: 513 associated, 34 explicitly labeled proxy contexts and 28 without a supported current source-area association. It publishes 526 distinct Native Resilience source areas and 4 separate USGS county/watershed contexts. Shared source areas are counted once in the climate table; no Nation-weighted national climate mean is inferred.

These areas support climate interpretation; they are not comprehensive sovereign territories. A proxy is neighboring or regional context, not a validated Nation mean. Several Nations may share a source area; Navajo agency areas remain separate choices. Geometry membership uses the documented public source-area intersection, not political region membership.

Native Resilience uses annual NASA NEX-GDDP-CMIP6 member values at approximately 25 km source resolution. Each model is compared with its own 1995-2014 baseline before equal-model ensemble summaries are calculated. Values below are 20-year mean changes; brackets show the 10th-90th percentile of model changes, not confidence intervals or scenario probabilities.

SSP2-4.5 and SSP3-7.0 are conditional pathways. Differences need not be monotonic at an individual place or time window. Temperature is degrees Celsius; precipitation is change in mean annual total, millimetres per year. Source annual tables do not retain realization IDs, calendar names or valid daily cell coverage.

The three story footprints provide regional context. Each example below uses one explicitly named climate source area selected by the explorer; it is not a climate average over the story footprint.

Columbia River Basin — U.S. portion

Selected place: **Confederated Tribes and Bands of the Yakama Nation**. Climate source area: **Yakama Nation Reservation (4690R)**; association: associated. Baseline: 1995-2014. Source family: Native Resilience / NASA NEX-GDDP-CMIP6.

Story membership rule: Union of listed authoritative units, clipped to 2025 U.S. county coverage; source-area membership requires more than 1 square metre positive intersection in EPSG:6933. Unique whole counties intersecting the story footprint; all county assets, not Nation assets or ecological-area loss.

Source: Native Resilience / NASA NEX-GDDP-CMIP6. Annual source SHA-256: be15b978a45c3aca7d84b61f32762c3a3cc10b148d9866041f034c981509dd21.

Metric / unit	20-year window	SSP2-4.5 median [10th, 90th]	SSP3-7.0 median [10th, 90th]	Models
Annual mean temperature (°C)	2020-2039	+0.89 [+0.59, +1.41]	+0.73 [+0.55, +1.34]	8 / 8
Annual mean temperature (°C)	2040-2059	+1.48 [+1.01, +1.80]	+1.52 [+1.25, +2.13]	8 / 8
Annual mean temperature (°C)	2070-2089	+2.11 [+1.78, +2.61]	+2.77 [+2.51, +3.39]	8 / 8
Annual precipitation (mm/year)	2020-2039	+15.89 [-8.95, +52.81]	+9.37 [-10.37, +54.66]	8 / 8
Annual precipitation (mm/year)	2040-2059	+40.31 [+18.08, +68.56]	+28.19 [+4.45, +83.86]	8 / 8
Annual precipitation (mm/year)	2070-2089	+44.60 [+18.95, +73.92]	+43.37 [+22.81, +65.30]	8 / 8

Desert Southwest

Selected place: **Gila River Indian Community of the Gila River Indian Reservation, Arizona**. Climate source area: **Gila River Indian Reservation (1310R)**; association: associated. Baseline: 1995-2014. Source family: Native Resilience / NASA NEX-GDDP-CMIP6.

Story membership rule: Union of listed authoritative units, clipped to 2025 U.S. county coverage; source-area membership requires more than 1 square metre positive intersection in EPSG:6933. Unique whole counties intersecting the story footprint; all county assets, not Nation assets or ecological-area loss.

Source: Native Resilience / NASA NEX-GDDP-CMIP6. Annual source SHA-256: ed9d3b4dd0f253736ab584118389de85258dbf8ddb57ab27c32056fd99a7fb5f.

Metric / unit	20-year window	SSP2-4.5 median [10th, 90th]	SSP3-7.0 median [10th, 90th]	Models
Annual mean temperature (°C)	2020-2039	+0.93 [+0.74, +1.27]	+0.95 [+0.68, +1.40]	8 / 8
Annual mean temperature (°C)	2040-2059	+1.85 [+1.24, +2.08]	+1.88 [+1.44, +2.20]	8 / 8
Annual mean temperature (°C)	2070-2089	+2.28 [+1.84, +2.76]	+3.17 [+2.74, +3.93]	8 / 8
Annual precipitation (mm/year)	2020-2039	+11.15 [-20.31, +37.27]	+3.34 [-40.44, +30.58]	8 / 8
Annual precipitation (mm/year)	2040-2059	+10.80 [-41.09, +50.58]	-0.14 [-37.51, +29.34]	8 / 8
Annual precipitation (mm/year)	2070-2089	-7.73 [-30.46, +10.76]	+8.28 [-73.88, +24.93]	8 / 8

Northeast forest

Selected place: **Penobscot Nation**. Climate source area: **Penobscot Reservation (2760R)**; association: associated. Baseline: 1995-2014. Source family: Native Resilience / NASA NEX-GDDP-CMIP6.

Story membership rule: Union of listed authoritative units, clipped to 2025 U.S. county coverage; source-area membership requires more than 1 square metre positive intersection in EPSG:6933. Unique whole counties intersecting the story footprint; all county assets, not Nation assets or ecological-area loss.

Source: Native Resilience / NASA NEX-GDDP-CMIP6. Annual source SHA-256: e4ededf2b48205da82b91396940587ca6f96829d88857bc4c885ca3f3d68bf03.

Metric / unit	20-year window	SSP2-4.5 median [10th, 90th]	SSP3-7.0 median [10th, 90th]	Models
Annual mean temperature (°C)	2020-2039	+1.15 [+0.77, +1.48]	+1.03 [+0.63, +1.61]	8 / 8
Annual mean temperature (°C)	2040-2059	+1.67 [+1.30, +2.33]	+1.72 [+1.22, +2.49]	8 / 8
Annual mean temperature (°C)	2070-2089	+2.63 [+2.16, +3.11]	+3.41 [+3.14, +4.19]	8 / 8
Annual precipitation (mm/year)	2020-2039	+65.48 [+28.00, +136.62]	+64.61 [+13.99, +95.01]	8 / 8
Annual precipitation (mm/year)	2040-2059	+83.56 [+58.03, +120.71]	+67.81 [+23.03, +120.03]	8 / 8
Annual precipitation (mm/year)	2070-2089	+103.12 [+66.05, +166.20]	+124.26 [+64.75, +189.37]	8 / 8

Separate USGS annual regional context

USGS NCCV LOCA2 uses a separate 1981-2010 baseline. These selected annual temperature anomalies (degrees Celsius) are copied from the provider summaries; they are not 20-year averages or year-specific weather forecasts. The explorer retains the full 1950-2100 series. SSP2-4.5 has 24 models and SSP3-7.0 has 23; memberships differ.

Prepared summaries are labeled October 2023 under the pinned 2026-02-03 publication path. That path does not establish incorporation of corrected 2024 precipitation. Regional precipitation is unavailable here. Annual provider quantiles have not been averaged into a new multi-year model spread.

Regional source area	Year	SSP2-4.5 annual anomaly [10th, 90th]	SSP3-7.0 annual anomaly [10th, 90th]	Models
Whatcom County, Washington (US53073)	2050	+2.08 [+0.87, +3.93]	+1.94 [+0.70, +2.91]	24 / 23
Whatcom County, Washington (US53073)	2100	+3.54 [+1.78, +5.62]	+4.59 [+3.18, +7.10]	24 / 23
Middle Columbia (HUC1707)	2050	+2.19 [+0.85, +3.27]	+2.14 [+1.00, +2.77]	24 / 23
Middle Columbia (HUC1707)	2100	+3.50 [+2.03, +5.47]	+4.61 [+3.43, +6.39]	24 / 23
Lower Colorado-Lake Mead (HUC1501)	2050	+2.33 [+1.49, +3.08]	+2.49 [+1.30, +3.11]	24 / 23
Lower Colorado-Lake Mead (HUC1501)	2100	+3.34 [+1.92, +5.18]	+4.91 [+3.54, +6.42]	24 / 23
Penobscot (HUC0102)	2050	+2.49 [+1.18, +4.01]	+2.22 [+1.34, +3.79]	24 / 23
Penobscot (HUC0102)	2100	+4.22 [+2.32, +5.33]	+6.01 [+4.56, +7.39]	24 / 23

County cost context for the three story footprints

Whole-county Expected Annual Loss covers all county assets and residents. It is an annualized model expectation, not Tribal loss, projected climate-change damage or funding need. Each county is counted once within a footprint; footprints can overlap and their totals must not be added. Missing fields remain unavailable.

FEMA National Risk Index December 2025 v1.20. The climate-sensitive subtotal uses the supported 15-hazard scope; it excludes earthquake, tsunami and volcano. A common price year is not supplied by the cached input. The companion CSV retains each hazard, county coverage and unmatched county IDs.

Story footprint	Matched / expected counties	Climate-sensitive EAL / year	All-hazard EAL / year
Columbia River Basin — U.S. portion	134 / 134	$3,988,220,939.03	$7,590,761,990.18
Desert Southwest	52 / 52	$12,002,002,163.91	$19,223,055,397.86
Northeast forest	174 / 174	$10,955,812,158.78	$11,472,637,806.04

Downloads, source versions and interpretation

Climate window values contains every supported public Native source-area window, metric and pathway with model counts and source hashes. Geography coverage accounts for all 575 Nation IDs. Cohort costs retains whole-county cost scope and missingness.

Native data terms: University of Montana Native Resilience projections, CC0. Current pinned producer commit: b8b55ca0e172a4694e4212108309621d8873615d. Native geometry vintage: 2025.

Regional source: Whatcom County, Washington; SHA-256 aaa6cf7d7f127c7692b4ba0bb713f85620f826645b078a0590235dc53da9f1b3.

Regional source: Middle Columbia; SHA-256 826cf3f54b4aafc135d0200f323ac3efd6e63450b56ab1b78b0f9913a7a93ed1.

Regional source: Lower Colorado-Lake Mead; SHA-256 c21fd8f691ea281e07c0c61bcbc5ce8475fd237a00a963bdd7d3f9333b427a7e.

Regional source: Penobscot; SHA-256 559694f573cc72349040744418266b740789fe10643c1698f3b85704e13f6b41.

Explorer release: catf-place-2026-10-22-v1. Catalog SHA-256: 14f99ea1d9e0c32720aaabfce87d146245e290897f47509ec0fbc3325d2d6bfc.

Public modeled land-area geometry is not a comprehensive statement of Nation territory, ownership or sovereign authority. Association carries a legacy reviewed name/code crosswalk forward to current source IDs; it is not independent Nation confirmation. Shared and borrowed areas remain explicit. EPA/CEC 2010 taxonomy does not cover Hawaii; the downloaded feature table has 182 Level III units; older linked metadata says 181. Geographic memberships may overlap. Deduplicate Nation IDs, source areas and county FIPS separately. County context includes assets outside Tribal lands and is not a future Tribal loss forecast. Display geometry is simplified separately; all intersections use unsimplified source geometry. Intersections at or below 1 square metre are numerical slivers and excluded. EPA/CEC boundaries are broad-scale ecological context at 1:250,000 source scale; small intersections do not establish locally precise ecological boundaries. Associated land areas carry forward the pinned crosswalk; the association is not an exhaustive inventory of each Nation’s lands.